

Event Management System

Problem Statement

Create a database to manage technical events, registrations, attendance, and certificates.

Modules

- Events
- Participants
- Registrations
- Attendance
- Certificates

Expected Outcome

Generate participant lists and attendance reports.

Database Design

- Events (event_id PK, event_name, event_type, event_date, venue, registration_fee)
- Participants (participant_id PK, participant_name, email, phone, city)
- Registrations (registration_id PK, event_id FK, participant_id FK, registration_date, registration_status)
- Attendance (attendance_id PK, registration_id FK, attendance_date, attendance_status)
- Certificates (certificate_id PK, registration_id FK, certificate_type, issue_date)

Sample Data / INSERT Queries

Events (8 Records)

INSERT INTO Events

(event_id, event_name, event_type, event_date, venue, registration_fee)

VALUES

(101, 'AI Summit 2026', 'Workshop', '2026-08-05', 'Hyderabad', 1500),
(102, 'Cloud Computing Expo', 'Conference', '2026-08-08', 'Bengaluru', 2000),
(103, 'Data Science Bootcamp', 'Workshop', '2026-08-12', 'Chennai', 1200),
(104, 'Cyber Security Meet', 'Seminar', '2026-08-15', 'Visakhapatnam', 800),
(105, 'Web Development Hackathon', 'Hackathon', '2026-08-18', 'Vijayawada', 500),
(106, 'Machine Learning Conference', 'Conference', '2026-08-20', 'Hyderabad', 1800),

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(107, 'DevOps Hands-on Lab', 'Workshop', '2026-08-23', 'Pune', 1000),
(108, 'Tech Career Connect', 'Seminar', '2026-08-25', 'Kochi', 600);
```

Participants (12 Records)

```
INSERT INTO Participants
(participant_id, participant_name, email, phone, city)
VALUES
(201, 'Aarav', 'aarav@gmail.com', '9876501001', 'Hyderabad'),
(202, 'Bhavya', 'bhavya@gmail.com', '9876501002', 'Vijayawada'),
(203, 'Charan', 'charan@gmail.com', '9876501003', 'Visakhapatnam'),
(204, 'Divya', 'divya@gmail.com', '9876501004', 'Bengaluru'),
(205, 'Esha', 'esha@gmail.com', '9876501005', 'Chennai'),
(206, 'Farhan', 'farhan@gmail.com', '9876501006', 'Mumbai'),
(207, 'Gopi', 'gopi@gmail.com', '9876501007', 'Warangal'),
(208, 'Harini', 'harini@gmail.com', '9876501008', 'Guntur'),
(209, 'Ishaan', 'ishaan@gmail.com', '9876501009', 'Pune'),
(210, 'Kavya', 'kavya@gmail.com', '9876501010', 'Kochi'),
(211, 'Manoj', 'manoj@gmail.com', '9876501011', 'Hyderabad'),
(212, 'Nandini', 'nandini@gmail.com', '9876501012', 'Bengaluru');
```

Registrations (16 Records)

```
INSERT INTO Registrations
(registration_id, event_id, participant_id, registration_date, registration_status)
VALUES
(301,101,201,'2026-07-20','Confirmed'),(302,101,202,'2026-07-21','Confirmed'),
(303,102,203,'2026-07-22','Confirmed'),(304,102,204,'2026-07-23','Cancelled'),
(305,103,205,'2026-07-24','Confirmed'),(306,103,206,'2026-07-25','Confirmed'),
(307,104,207,'2026-07-26','Confirmed'),(308,104,208,'2026-07-27','Confirmed'),
(309,105,209,'2026-07-28','Confirmed'),(310,106,210,'2026-07-29','Confirmed'),
(311,106,201,'2026-07-30','Confirmed'),(312,107,211,'2026-07-31','Confirmed'),
(313,107,212,'2026-08-01','Cancelled'),(314,108,203,'2026-08-02','Confirmed'),
(315,103,201,'2026-08-03','Confirmed'),(316,105,205,'2026-08-04','Confirmed');
```

Attendance (14 Records)

```
INSERT INTO Attendance
(attendance_id, registration_id, attendance_date, attendance_status)
VALUES
(401,301,'2026-08-05','Present'),(402,302,'2026-08-05','Present'),
(403,303,'2026-08-08','Present'),(404,305,'2026-08-12','Absent'),
(405,306,'2026-08-12','Present'),(406,307,'2026-08-15','Present'),
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(407,308,'2026-08-15','Absent'),(408,309,'2026-08-18','Present'),  
(409,310,'2026-08-20','Present'),(410,311,'2026-08-20','Present'),  
(411,312,'2026-08-23','Present'),(412,314,'2026-08-25','Absent'),  
(413,315,'2026-08-12','Present'),(414,316,'2026-08-18','Present');
```

Certificates (11 Records)

```
INSERT INTO Certificates  
(certificate_id, registration_id, certificate_type, issue_date)  
VALUES  
(501,301,'Participation','2026-08-05'),(502,302,'Participation','2026-08-05'),  
(503,303,'Participation','2026-08-08'),(504,306,'Participation','2026-08-12'),  
(505,307,'Participation','2026-08-15'),(506,309,'Winner','2026-08-18'),  
(507,310,'Participation','2026-08-20'),(508,311,'Participation','2026-08-20'),  
(509,312,'Participation','2026-08-23'),(510,314,'Participation','2026-08-25'),  
(511,315,'Participation','2026-08-12');
```

SQL Questions

1. Display all events along with their event type and venue.
2. Display the names and email addresses of participants from a particular city.
3. Update the registration fee of all workshops by 10%.
4. Change the registration status from Confirmed to Cancelled for a selected registration.
5. Delete a cancelled registration.
6. Display all participant names in uppercase.
7. Display event names in lowercase.
8. Display the first four characters of each event name.
9. Concatenate the participant's name and city into a single column.
10. Replace the word 'Conference' with 'Summit' in the event type column.
11. Display events scheduled in the current month.
12. Display the day name for each event date.
13. Calculate the number of days between the registration date and event date.
14. Display event dates in the format DD-Month-YYYY.
15. Display the month and year of each certificate issue date.
16. Count the number of participants registered for each event.
17. Find the average registration fee for each event type.
18. Find the highest registration fee among all events.
19. Find the total number of registrations for each registration status.
20. Count the number of Present and Absent participants for each event.
21. Display event types having more than one event.

22. Find the total number of certificates issued for each certificate type.
23. Display participant names along with the events they registered for.
24. Display event names, participant names, and registration dates.
25. Display participant names, event names, attendance status, and attendance date.
26. Display participant names and the certificates they received.
27. Display event names along with their registered participants, including events with no registrations.
28. Display all participants and their registration details, including participants who have not registered for any event.
29. Display event name, participant name, registration status, and certificate type in one result.
30. Find participants who registered for events with a registration fee greater than the average event registration fee.
31. Find the event with the highest number of registrations.
32. Display participants who attended more events than the average attendance count per participant.
33. Find participants who have attended at least one workshop.
34. Display events that have never received a registration.
35. Create a view named Participant_Event_Details displaying participant name, event name, event date, registration status, and attendance status.
36. Create a view named Event_Attendance_Summary displaying event name, total registrations, total present, and total absent. Retrieve records from both views.
37. Start a transaction. Update the registration fee of an event. Create a SAVEPOINT. Update a registration status. Roll back to the SAVEPOINT and then commit the transaction.
38. Create a stored procedure to display all registrations made by a particular participant.
 - Input: participant_id
 - Display: registration_id, event name, event date, registration date, and registration status.
39. Create a stored procedure to calculate the total number of events attended by particular participant.
 - Input: participant_id
 - Use the Attendance table and display the total number of Present records.
40. Create a trigger on the Registrations table. Whenever a registration is deleted, automatically store the deleted registration details in a Registrations_Archive table.
41. Create a trigger on the Attendance table. Whenever the attendance status changes, automatically store the registration ID, old status, new status, and change date in an Attendance_History table.